

Data Network Services

Private Consulting Engagement

Scope of Work

Overview: Pursuant to a review and write up of issues occurring on the Brecksville Broadview Heights City School Districts LAN/WAN serving the current 6 school buildings, transportation building and the Board of Education building, TTG is supplying this SOW to perform the remediation as describe and outlined in that report.

TTG Scope: TTG will provide a network architecture expert to redesign the network routing and architecture currently serving the district to include separate and distinct VLANs for Data/Voice/Video/Internet Traffic. This will include but not be limited to the following:

- Ruckus network equipment validation completed, no equipment problems noted
 - Scale of devices
 - Sizing and ports
 - Interface sizing, error count checks, and utilization
 - Processor and utilization
- Layer 2 switching structures validated, implementation completed but overall network design problem
 - Spanning tree per vlan in good order
 - Leaf node path count high but not critical
 - Root bridge priority set and implemented correctly
 - Some vlans are spanned across all sites including WAN sites producing latency and collisions. Needs Corrected
 - Massive broadcast domains of end user devices. Needs Attention
- Layer 3 routing validated
 - Default routing noted through bridge devices then to Connect provider.
 - Once packet reaches Connect no further information is known
 - Only one building with layer 3 interfaces all buildings must span to reach for routing between vlans or outbound to Internet.
 - Routing design mixed with Connect provider

Time Frame: This work is going to involve a significant amount of time in planning and design, implementation, and documentation. The current outline for services calls for up to 500 hours of our Director of Engineering time and efforts. TTG understands that the need for the district to implement it new VoIP phone system and can commit to completing its work on this network in no more than 12 weeks from the kick off meeting with BBHCS. TTG will be devoting its director

to Brecksville 3 days (onsite and remote) per week at the onset of the project for the design, routing, and implementation of the project. We will commit this level of involvement until the project is complete and for all weeks except holiday weeks which will be determined with the district.

Deliverable: The following deliverables will accompany this project:

- Weekly summary of work accomplished and Brecksville IT direction for following week
- IP Documentation
- Design overview complete with VLAN mapping and documentation
- Regular project overview for upcoming plan implementations and changes designed to minimize downtime, prevent outages, and increase back in-service times for any outages that may occur
- Final network documentation

Cost: TTG's cost for this SOW work will not exceed \$76,000 and includes everything necessary to complete the redesign and implementation. TTG will secure a network engineer to assist in the deployment of this redesign. If upon successful completion of the network redesign SOW outlined the District believes it is in their best interest to present an offer of employment to the network design engineer it may do so without the incurrence of any reparation demands or charges from TTG. The network engineer shall be free to accept or decline the offer based on its merits. Should the district find and hire a network engineer while the project is still in progress, TTG will work with the district to restructure the work and the remaining fees based on where we are in the project and the capabilities of the new engineer. TTG will share all project related information with the school district.

TTG shall bill the district in three installments; \$10,000 at to outset of the project, \$36,000 upon the delivery and review of the design and programming documentation, and \$30,000 upon the delivery of final network documentation and implementation.

This represents the entire agreement for services and scope of work. Work outside of the agreement may be identified and will not be started without district approval and signoff on an additional SOW.

Telecommunication Technologies Group LLC

Brecksville Broadview Heights City School District

Date

Date